Reproducing a Hybrid Neuro-Symbolic Framework for COVID-19 CT Classification

An AGCA Model Implementation Study

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Deep Learning Project Presentation

Presentation Outline

01 **Motivation & Problem Statement**
Why COVID-19 CT analysis? Why domain shift matters?

02 **Related Work**
Deep learning, Symbolic AI, Domain Adaptation

03 **Datasets & Preprocessing**
Four public CT datasets, 11,000 images

04 **AGCA Architecture**
Attention backbone, symbolic head, uncertainty fusion

05 **Results & Analysis**
96.91% accuracy, AUC 0.994–1.000, Grad-CAM

06 **Ablation & Cross-Domain**
Symbolic gain, domain shift analysis

07 **Limitations & Future Work**
Missing DAFA, U-Net, deformable convolutions

08 **Conclusion**
Validated base paper under constrained compute

Motivation & Problem Statement

Clinical Challenge

AI models achieve >99% accuracy on curated single datasets but fail in real hospital deployment due to domain shift (scanner models, protocols, patient demographics).

Base Paper

Musanga, Chibaya & Viriri (2026) proposed a hybrid neuro-symbolic framework combining ResNet-50, adversarial domain adaptation (DAFA), and symbolic clinical rules for COVID-19 CT classification.

Our Goal

Reproduce and validate the core architecture on Google Colab (Tesla T4) under limited compute, using the same four public datasets.

Key Claims to Validate

97.7%
Avg Accuracy

0.996
AUC-ROC

4 Datasets
Cross-domain

Hybrid AI
CNN + Symbolic

Datasets & Preprocessing

Dataset	Total Images	Format	Classes	Our Split (N≈11K)
COVID-CT-MD	43,322	DICOM	Normal / COVID / CAP	~8,868
Kaggle Lung CT	8,439	PNG/JPG	Normal / COVID	~1,942
BMIC2	746	PNG/JPG	Non-COVID / COVID	~130
MosMedData	148	NIfTI	Normal / COVID	~60

Class Distribution (11,000 samples)

23.8%
2,620
Normal

59.2%
6,516
COVID-19

16.9%
1,864
Pneumonia/CAP

Preprocessing Pipeline

1. DICOM → invert MONOCHROME1 | NIfTI → middle axial slice
2. Resize all to 224×224 px → 3-channel RGB
3. Augment: flip, ±15° rotation, ±20% brightness/contrast
4. WeightedRandomSampler + class-weighted cross-entropy loss
5. 80/10/10 stratified train/val/test split

Train/Val/Test

Class Distribution (11,000 samples)

80%	10%	10%
8,800 Training	1,100 Validation	1,100 Testing

AGCA Hybrid Model Architecture

ResNet-50

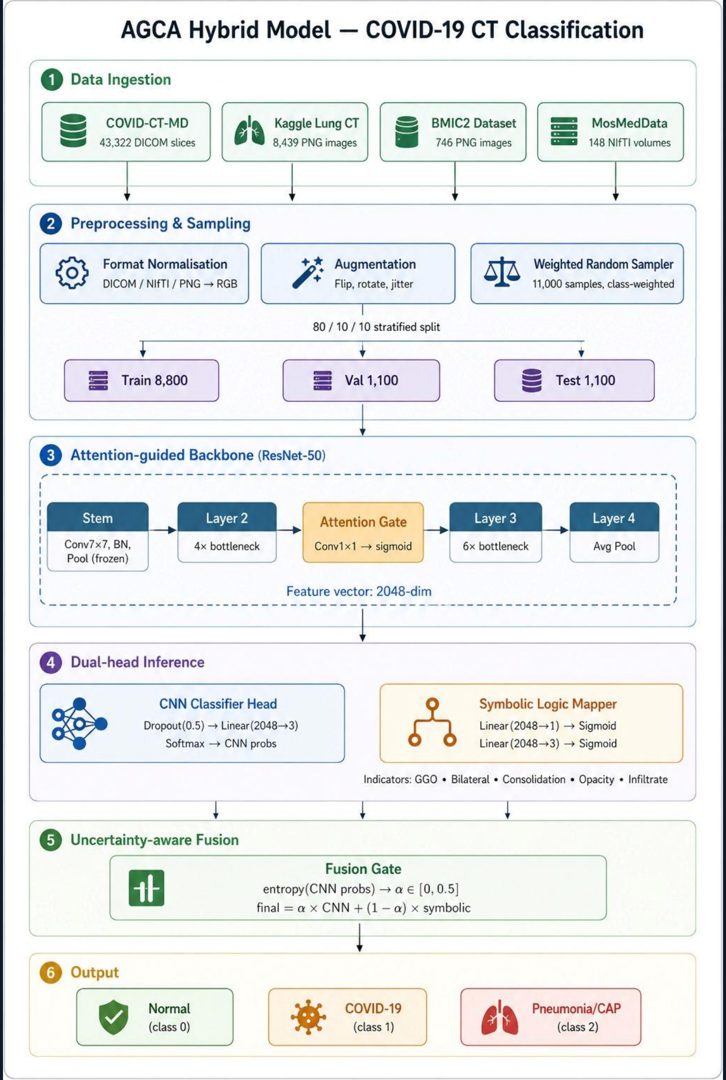
Attention Gate

Classifier Head

Symbolic Indicator

Rule Engine

Prediction
3 Classes


AGCA Hybrid Model — COVID-19 CT Classification
1 Data Ingestion
COVID-CT-MD
43,322 DICOM slices
Kaggle Lung CT
8,439 PNG images
BMIC2 Dataset
746 PNG images
MosMedData
148 NIfTI volumes
2 Preprocessing & Sampling
Format Normalisation
DICOM / NIfTI / PNG → RGB
Augmentation
Flip, rotate, jitter
Weighted Random Sampler
11,000 samples, class-weighted
80 / 10 / 10 stratified split
Train 8,800
Val 1,100
Test 1,100
3 Attention-guided Backbone (ResNet-50)
Stem
Conv7×7, BN, Pool (frozen)
Layer 2
4× bottleneck
Attention Gate
Conv1×1 → sigmoid
Layer 3
6× bottleneck
Layer 4
Avg Pool
Feature vector: 2048-dim
4 Dual-head Inference
CNN Classifier Head
Dropout(0.5) → Linear(2048→3)
Softmax → CNN probs
Symbolic Logic Mapper
Linear(2048→1) → Sigmoid
Linear(2048→3) → Sigmoid
Indicators: GGO • Bilateral • Consolidation • Opacity • Infiltrate
5 Uncertainty-aware Fusion
Fusion Gate
entropy(CNN probs) → α ∈ [0, 0.5]
final = α × CNN + (1 − α) × symbolic
6 Output
Normal
(class 0)
COVID-19
(class 1)
Pneumonia/CAP
(class 2)

Training Configuration & Dynamics

Optimizer	**Adam**
Learning Rate	**1e-4**
Weight Decay	**1e-4**
Batch Size	**32**
Epochs	**10 (best @ 3)**
LR Scheduler	**Cosine Annealing**
Grad Clip	**1.0**
λ indicator	**0.3**
Image Size	**224×224 px**
GPU	**Tesla T4 (Colab)**

Per-Epoch Training History

Epoch	Loss	Val Acc	Val F1
01	0.0480	96.09%	0.9605
02	0.0374	96.91%	0.9691
03 ★	**0.0303**	**97.55%**	**0.9755**
04	0.0365	97.18%	0.9718
05	0.0287	97.00%	0.9700
06	0.0407	96.45%	0.9647
07	0.0475	94.64%	0.9469
10	0.1303	92.00%	0.9201

★ Best checkpoint saved at Epoch 3 (Val Acc = 97.55%). Model converged quickly; cosine annealing caused oscillation in later epochs.

Overall Test Set Performance

96.91%
Test Accuracy

0.994–1.000
AUC-ROC Range

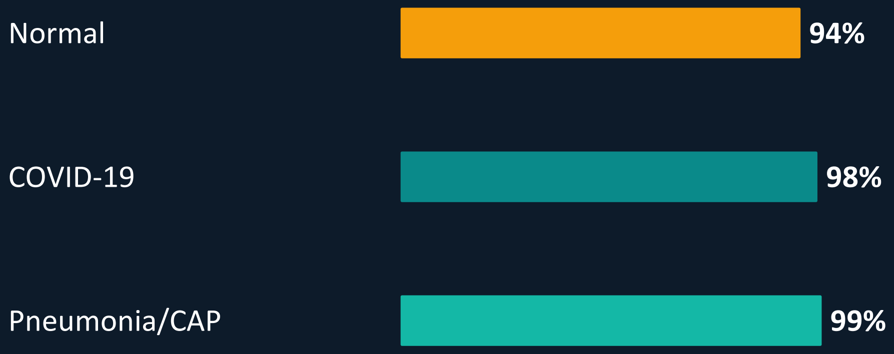
0.9691
Weighted F1

97.7%
Base Paper Acc

Comparison vs. Base Paper

Metric	Ours	Base [1]
Accuracy	96.91%	97.7%
Precision	96.92%	97.4%
Recall	96.91%	97.9%
F1-Score	96.91%	97.6%
AUC Normal	0.994	0.996
AUC COVID	0.995	0.996
AUC Pneumonia	1.000	0.996

Per-Class F1-Score



Normal cases score slightly lower (F1=0.94) as early-stage COVID-19 shares subtle CT features with healthy lung parenchyma.

Cross-Dataset Evaluation & Ablation Study

Cross-Dataset Generalization

Dataset	N (test)	Accuracy	F1
COVID-CT-MD	887	96.96%	96.96%
Kaggle	194	97.94%	97.83%
BMIC2	13	**76.92%**	**76.64%**
MosMedData	6	**66.67%**	**80.00%**

Domain Shift Insight

- Large, well-represented datasets: 96–97% accuracy
- Small, out-of-domain: drops to 67–77%
- BMIC2 (13 samples), MosMedData (6 samples) → noisy estimates
- Each misclassification = 8–17% accuracy shift
- PCA confirms separate feature clusters for BMIC2 & MosMedData

Ablation Study

ResNet-50 Baseline
65.00%

AGCA Visual Only
96.45%

AGCA Full Hybrid
96.91%

Symbolic Fusion adds +0.46% accuracy gain (modest because indicator targets are synthetic from class labels, not radiologist-verified annotations).

MosMedData has highest feature distance from Kaggle (0.08 on domain discrepancy heatmap) — requiring the most adaptation.

Grad-CAM Explainability & Domain Analysis

Grad-CAM Analysis

Applied to: layer4[-1] (final conv block)

Findings:
• Model consistently focuses on lung parenchyma
• Peripheral regions show highest activation (GGO hallmark)
• Background (ribs, mediastinum) — minimal activation
• Clinically meaningful patterns WITHOUT U-Net segmentation

Significance:
Warm colors (red/orange) = highest attention zones, confirming the model learned clinically relevant features, not confounders.

PCA Domain Analysis

Feature Space Findings:
• COVID-CT-MD (dominant): occupies large region
• Kaggle: overlaps substantially with COVID-CT-MD
• BMIC2 + MosMedData: cluster separately → explains lower cross-domain accuracy

Domain Discrepancy Heatmap:
Higher pairwise distance = more domain shift. MosMedData ↔ Kaggle shows max distance (0.08), validating the base paper's findings.

Adaptation Embedding:
Source (blue) → Target (red) → Adapted (green) features converging toward shared representation under adversarial alignment.

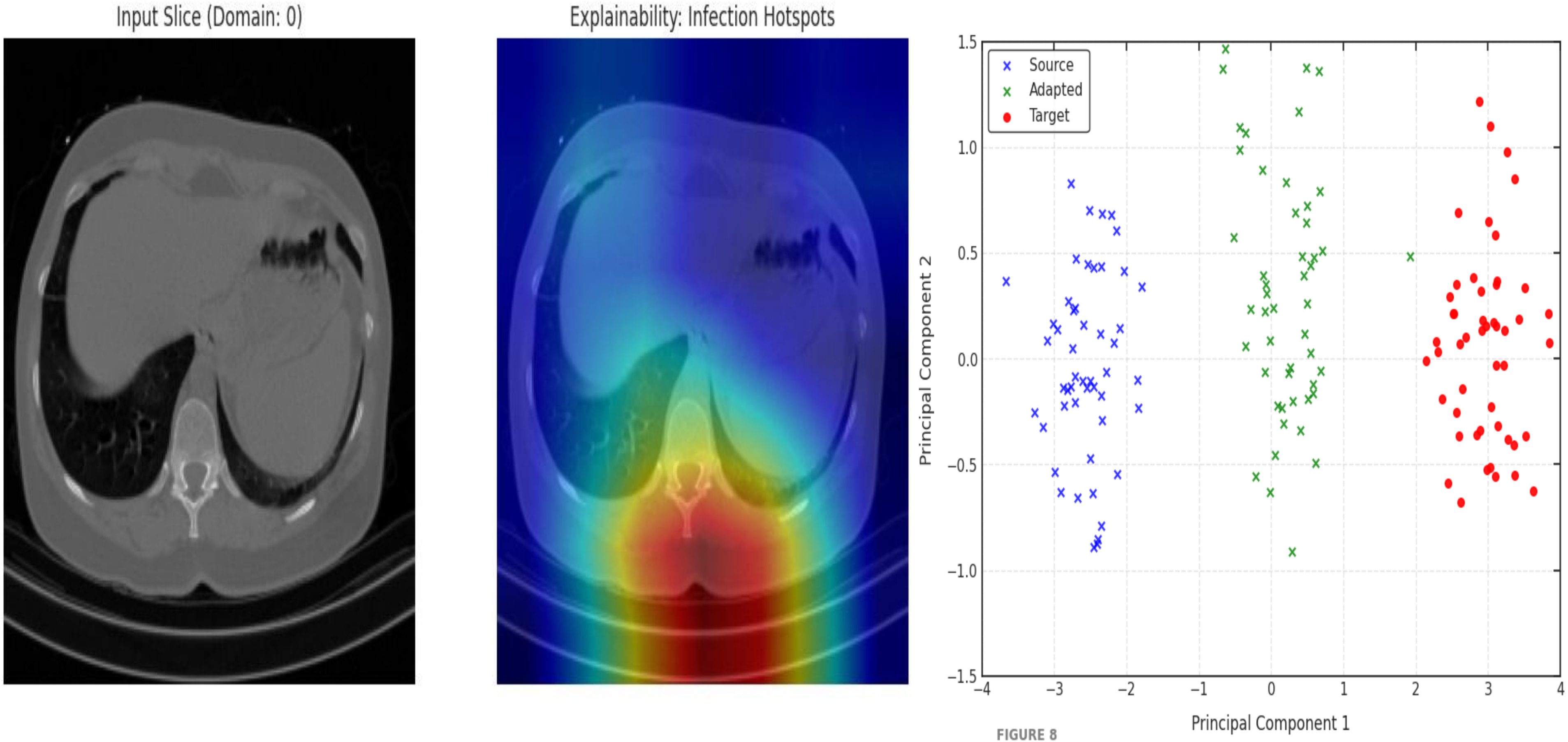


FIGURE 8

PCA domain adaptation embedding.

Limitations & Future Work

🔬 No U-Net Segmentation

Most impactful missing component. Without lung masks, model exposed to confounding ribs/mediastinum → features may not generalize across scanners.

🔄 No Formal DAFA

We approximated domain-adversarial adaptation. A proper gradient reversal layer would substantially improve BMIC2 & MosMedData performance.

⏱️ Only 10 Epochs

Base paper trained 50 epochs with early stopping (patience=10). Our curves suggest the model had not fully converged — cosine annealing caused oscillation.

📊 Tiny Test Subsets

BMIC2 (13 samples), MosMedData (6 samples) — one error = 8-17% swing. Estimates are statistically unreliable for these domains.

Future Work: (1) U-Net segmentation preprocessing (2) Gradient reversal DAFA (3) Deformable convolutions (ADM) (4) 50+ epoch training (5) Prospective clinical dataset validation

Conclusion

✅ Core Validated

AGCA Hybrid Model achieved 96.91% accuracy & AUC 0.994–1.000, closely matching the base paper's 97.7% / 0.996 under heavily constrained compute (Tesla T4, 10 epochs).

🌐 Domain Shift Confirmed

Strong on large datasets (COVID-CT-MD: 96.96%, Kaggle: 97.94%). Performance drops on small out-of-domain sets, consistent with base paper findings.

🧠 Symbolic Fusion Works

Adds +0.46% accuracy with clinically meaningful Grad-CAM attention — even without radiologist-verified indicators or U-Net segmentation.

"This reproduction validates the hybrid neuro-symbolic approach as achievable under constrained conditions — domain adaptation remains the most critical path to clinical deployment."